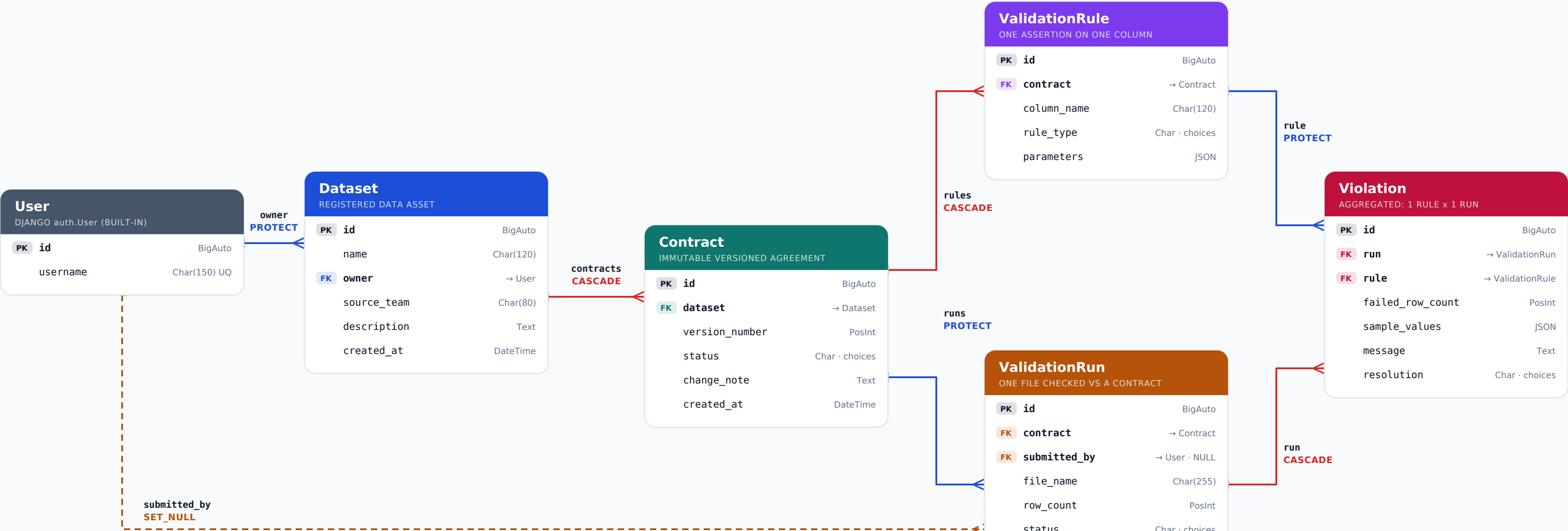


DataPact — Entity Relationship Diagram

INFO 490 · Project 1 · Assignment A1 | Team 15 — Hriday Agarwal, Ashok Chacko, Connor Slattery, Tejas Jaggi

5 application models + Django auth.User

Crow's-foot notation · every relationship is one-to-many



LEGEND — on_delete

CASCADE — child cannot exist alone

PROTECT — blocks deletion of the parent

SET_NULL — audit row outlives the user

| = exactly one > = zero or more

UNIQUE CONSTRAINTS

Dataset (owner, name)

a team cannot register the same asset twice

Contract (dataset, version_number)

version numbers stay dense and unambiguous

ValidationRule (contract, column_name, rule_type)

no duplicate assertions in a contract

Violation (run, rule)

one aggregated result per rule per run

DESIGN DECISIONS

A Contract is itself a version. Editing a contract that has already been used creates a new version rather than overwriting history, so a past run always points at the exact rules it was judged by.

ValidationRun reaches its Dataset through Contract → Dataset. No redundant dataset_id is stored on the run, so the two join paths can never disagree.

Violation holds one aggregated row per rule per run, not one row per failing cell.

PROTECT on Contract and ValidationRule makes the audit trail non-destructible by design.